Namal University, Mianwali

Project Proposal Document

Project: Point-of-Sale System

Course: CSC-225 – Software Engineering

Department: Computer Science

Semester: Fall 2025

Submitted To:
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Lecturer, Department of Computer Science

Submitted By:

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Submission Date: 9 Nov 2015

Requirement Provider Agreement

Project Title: Point-of-Sale System
Course: CSC-225 – Software Engineering
Department: Computer Science, Namal University

This agreement is made between the following parties.

Student Team:

Muhammad Shoaib	Reg No: Num-Bscs-2024-57
Allah Ditta	Reg No: Num-Bscs-2024-10
Zainab Bibi	Reg No: Num-Bscs-2024-79

Requirement Provider (RP):

Dr Muzamil Ahmed
Assistant Professor, Department of Computer Science, Namal University

Agreement Purpose:

The student team and the Requirement Provider agree to work together for the Software Engineering Project. Dr Muzamil will provide requirements, feedback, and guidance. The team will do regular meetings and share progress updates.

Muhammad Shoaib (Team Leader)	Signature: Shoaib
Dr Muzamil Ahmed (RP)	Signature: 7/11/25
Date: 7 Nov 2025	Date: 7 Nov 2025

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1 Introduction

Restaurants, such as pizza shops, face difficulties in managing orders, sales, and inventory when operating multiple branches.which is where a Point-of-Sale (POS) system can help.A Point-of-Sale (POS) system makes the running of restaurants much easier.It helps in taking orders, handling payments, and keeping everything organized even in busy hours.Shows menu and stock in one place, so even if different branches have different menus, the system shows the correct items for each branch.It also helps to track which dishes sell well, alerts staff when ingredients are running low, and ensures that prices are consistent across all branches,reducing mistakes, and making service faster and smoother.

2 Problem Statement

Mostly, the pizza shop manually handles all their managements.which often creates several problems.Orders are usually written by hand, leading to mistakes such as wrong items, missing custom requests or miscommunication between waiter and kitchen staff, a worker may take an order for a pizza that exists at another branch but not at their own if the restaurant has different branches. Prices or ingredients might also different and without a proper system its hard for staff to remember all those details.Manual billing and order tracking take more time,especially during busy hours,leading delay in service.Without a POS system its also difficult to get accurate sales reports, identify the best selling items,or monitoring staff performance.Customer may even miss out better deals or discount is their order is similar in value to a deal but is entered separately at a higher price.

3 Project Objectives

The objectives below are specific to the tasks assigned to the author and are written to be measurable and verifiable where possible.

1. To design a multi-branch compatible POS system for pizza shops.
2. To allow the admin to create, update, and delete menu items for each branch.

3. To track both cooked and raw inventory in real-time.
4. To provide automatic deal recommendations based on user selections.
5. To ensure efficient order processing and accurate billing.

4 Stakeholder Identification

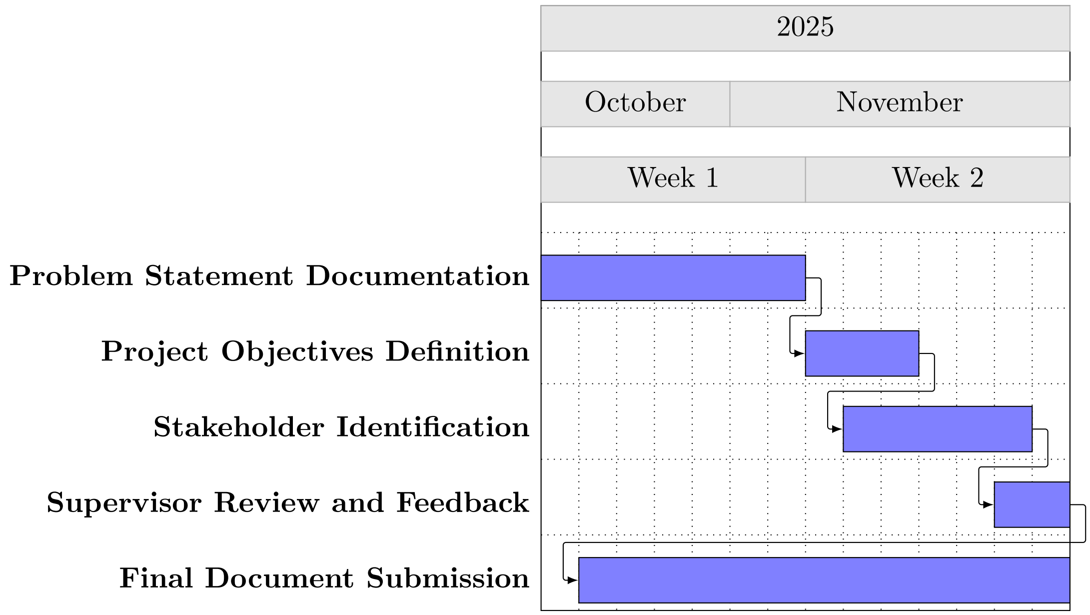
The stakeholders listed below are identified according to the project scope and the Requirement Engineering slides. For each stakeholder, we provide the role, responsibilities, and interests.

Stakeholder	**Role / Title**	**Responsibilities and Interests**
Requirement Provider	Client / End-user Representative	Provides requirements, validates functionality, reviews and signs off SRS, and provides feedback during development.
Administrator	System Owner / Head Office	Manages branches, monitors consolidated sales and inventory, creates and updates deals, configures branch policies, and reviews reports.
Branch Manager	Branch-level Supervisor	Operates branch-level management tasks: views and manages orders, updates local menu and availability, monitors branch inventory levels, and resolves customer issues.
Cashier / Staff	POS Terminal User	Processes in-shop orders, records payments, updates order status, and assists in inventory stock updates for sold inventory items.
Customer	End User	Places orders online or in-person, uses deal recommendations, views order status, and provides feedback.

Stakeholder	Role / Title	Responsibilities and Interests
Development Team	Designers and Implementers	Responsible for requirement analysis, SRS preparation, system design, implementation, testing, deployment, and maintenance.
Payment Gateway Provider	External System	Provides card / online payment processing and transaction settlement services.
Printer / Hardware Vendor	External Supplier	Supplies receipt printers, POS tablets, scanners, and supports hardware integration and maintenance.

5 Gantt Chart

Task	Start	End	Duration (Days)	Dependency
Problem Statement Documentation	10/27/2025	11/2/2025	7	-
Project Objectives Definition	11/3/2025	11/5/2025	7	Problem Statement
Stakeholder Identification	11/4/2025	11/8/2025	4	Project Objectives
Supervisor Review and Feedback	11/8/2025	11/9/2025	2	All previous tasks
Final Document Submission	10/28/2025	11/9/2025	2	Supervisor Review



6 Methodology

This project uses the Agile Scrum methodology. In Scrum, work is divided into short time periods called **sprints**. Each sprint focuses on completing a specific set of tasks and goals. At the end of each sprint, progress is reviewed and feedback is collected from team members or users. This helps the team understand what is going well and what needs improvement before starting the next sprint. It is one of the most popular and flexible software development methods.

6.1 Main Sprints of This Project

- Understanding user and system requirements
- Creating use case diagrams
- Designing the software interface
- Final documentation and making the prototype

Agile Scrum also includes regular meetings such as the daily **stand-up meeting**, where team members share updates on their work. This keeps everyone on the same page and helps resolve any issues quickly. Scrum is particularly useful when project requirements may change over time, as it allows updates or improvements without affecting the entire system. Users are involved throughout the project, ensuring the final product aligns with real user needs. Using Agile Scrum makes the project organized, interactive, easier to manage, saves time, improves teamwork, and helps deliver a better-quality prototype.

7 Tools and Technologies

In this project, the main focus is on understanding requirements and creating a working prototype. Several tools and technologies are used to organize and manage tasks effectively:

1. **Designing Tool: Figma**
   Figma is used for designing the layout and interface of the software, creating wireframes, and interactive prototypes. It allows easy sharing and feedback collection during development.

2. **Documentation Tool: LaTeX**
   LaTeX is used for writing and formatting the project report, organizing sections, headings, and references professionally.

3. **Version Control: GitHub**
   GitHub stores all project files, including reports, diagrams, and design documents. It provides version control so that changes are tracked and teamwork is easier to coordinate.

4. **Diagram Tools: Draw.io or Lucidchart**
   These tools are used to create Use Case Diagrams, Data Flow Diagrams (DFD), and other visual representations, helping clarify system requirements and design.

8 AI Tools Usage

While preparing this proposal, AI tools, especially ChatGPT (by OpenAI), were used to assist in various areas like writing ideas, improving clarity, and formatting. The AI tools were used for:

- Generating ideas for project features and proposed system
- Improving clarity, grammar, and structure of written content
- Researching best practices for software development methodologies
- Suggesting appropriate tools and technologies for the project

ChatGPT helped explain technical parts in simpler terms, fix some LaTeX formatting issues, and give ideas on organizing the report. All AI-generated content was reviewed, modified, and approved by the project team. Final decisions, content approval, and design implementation were done solely by the team members.

Reference

OpenAI. (2025). *ChatGPT [Large language model].* Retrieved from `https://chat.openai.com`